2. Forecasting and Valuing Cash Flows

MGMT 648: Valuation

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Today's Session

Defining Investment Cash Flows

- Only incremental cash flows count
- Sunk costs and cannibalization
- Conservative vs. optimistic forecasts
- Project FCF vs. equity FCF

Calculating Free Cash Flow

- The FCF formula
- NOPAT, depreciation, CAPEX
- Net working capital
- Comprehensive example: Lecion Electronics

Valuing Cash Flows

- Discounting FCF to present value
- Net present value (NPV)
- Internal rate of return (IRR)
- Mutually exclusive projects

The DCF Framework

The Three-Step Process

Step	Question	Tool
1	How much cash is the project expected to generate and when?	Forecast free cash flow (FCF)
2	How risky are the future cash flows? What return do investors require?	Weighted average cost of capital (WACC)
3	What is the present value of the investment's expected future cash flows?	Discount FCF using WACC

> This chapter focuses entirely on **Step 1** — the "heavy lifting" of DCF analysis. Step 2 (discount rates) is covered in Chapters 4 and 5.

Step 1 is the Hard Part

Why Forecasting Is Difficult

- The future can never be known with certainty
- Forecasts often must be made with little or no historical data
- The further out you forecast, the more uncertainty compounds
- Even experts get it badly wrong — Thomas Watson (IBM, 1943): “I think there is a world market for maybe five computers”

The Good News

- Good forecasting is part science, part art, part experience
- A rigorous framework forces you to make your assumptions explicit
- Explicit assumptions can be tested, challenged, and updated
- Sensitivity analysis reveals which assumptions actually drive the value

The point is not to achieve perfect forecasts — it is to build forecasts that are grounded in sound economic

Defining Investment Cash Flows

Three Key Questions

Before building any financial model, get these three questions right:

1. What cash flows are relevant?

Not all cash flows associated with an investment actually belong in its valuation.

Only **incremental** cash flows — those that would not occur if the project is rejected — are relevant.

2. Are the forecasts expected or hoped-for?

Theory calls for **expected** cash flows: probability-weighted averages across all scenarios.

In practice, forecasts are often optimistic projections of what happens "if all goes as planned."

3. Project FCF or equity FCF?

Project FCF is the cash available to all investors — debt and equity combined.

Equity FCF deducts interest payments and debt repayments. The choice determines which discount rate to use.

An incremental cash flow is one that occurs **only if the project is accepted**. If it happens regardless of the decision, it is not incremental.

Direct Effects

The revenues and costs generated by the project itself — the obvious ones.

Example: a new product line's sales, cost of goods sold, and operating expenses.

Indirect Effects

Effects the project has on the firm's *other* lines of business.

Example: when Frito-Lay launches a new Doritos flavor, the new flavor's sales may cannibalize existing Doritos varieties. The lost sales are a real cost of the new product.

The Cannibalization Problem

Frito-Lay's Approach to Incremental Revenue

Frito-Lay (a PepsiCo division) classifies new product offerings into three types based on cannibalization risk:

- **High cannibalization risk** (e.g., a new Doritos flavor): only a small fraction of revenue is truly incremental — the rest comes from lost sales of existing products
- **Moderate cannibalization risk** (e.g., baked chips): these open new shelf space and provide some real benefit, so a higher fraction is counted as incremental
- **Low cannibalization risk** (e.g., Frito-Lay's Natural line): these enter new channels with minimal overlap — the highest percentage of revenue is treated as incremental

The key discipline: estimate total revenue, then estimate what percentage is **truly new to the firm**. The rest is just revenue shifting — not value creation.

What Is a Sunk Cost?

A cost that has **already been incurred** — or must be incurred regardless of the investment decision.

Sunk costs cannot be recovered. The go/no-go decision cannot change them. Therefore they have no bearing on whether to proceed.

Common examples: past R&D spending, market research, feasibility studies, consultant fees paid to evaluate the opportunity.

The Merck Example

Merck spent $10M developing a stem cell methodology. Clinical trials were disappointing. A new application for cattle emerged.

To develop the cattle application, Merck must forgo a $8M licensing offer from Pfizer.

The relevant cost of proceeding: **$8M** (the opportunity cost) — not $10M. The $10M is gone either way. Only the $8M is affected by the current decision.

Conservative vs. Optimistic Forecasts

What Theory Assumes

DCF theory calls for **expected cash flows** — probability-weighted averages across all scenarios.

If a project generates $50K or $100K with equal probability, the expected cash flow is $75K. Discount $75K at the risk-adjusted rate.

What Practice Delivers

Managers often use **hoped-for** cash flows — “if all goes as planned.”

These optimistic forecasts ignore unanticipated glitches and assume best-case outcomes.

When hope-based forecasts are unavoidable (e.g., emerging market ventures, startup investments), use a **higher discount rate** to adjust for the gap between dreams and reality.

Incentive bias and overconfidence both push forecasts upward. The Phase II review process exists partly to

Project FCF vs. Equity FCF

Project (Firm) FCF

Cash available to **all investors** — both creditors and equity holders.

- Taxes computed on EBIT (before interest)
- Interest expense is NOT deducted
- Discount using WACC (blended cost of all capital)

Use this for standard project and enterprise valuation.

Equity FCF

Cash available to **equity holders only**.

- Project FCF minus after-tax interest payments
- Plus net new debt issued
- Discount using the cost of equity

Use this when you want to value just the equity stake (e.g., calculating stock price directly).

This course focuses primarily on **project FCF**. We will use equity FCF later when valuing businesses from the

Calculating Free Cash Flow

The FCF Formula

$$\text{FCF} = \underbrace{\text{EBIT} \times (1 - T)}_{\text{NOPAT}} + \text{Depreciation} - \text{CAPEX} - \Delta\text{Net Working Capital}$$

Line Item	Direction	Why
EBIT × (1 − T) = NOPAT	Starting point	Operating profit after taxes, before financing
+ Depreciation	Add back	Non-cash charge — reduces taxes but not cash
− Capital Expenditures	Subtract	Real cash outflow for plant and equipment
− Change in Net Working Capital	Subtract	Cash tied up in receivables and inventory
= **Free Cash Flow**	Result	Cash available to all investors

NOPAT: Net Operating Profit After Taxes

Why Start with EBIT?

Project FCF is the cash available to **all capital providers** — debt and equity combined.

To get there, we compute taxes on EBIT (before interest), rather than on EBT (after interest). This keeps financing costs out of the operating cash flow calculation.

The discount rate (WACC) separately accounts for the tax benefit of debt.

NOPAT in Practice

NOPAT = EBIT × (1 − Tax Rate)

Example: EBIT = $100K, tax rate = 30%

NOPAT = $100K × (1 − 0.30) = $70K

Then add back depreciation: if depreciation is $40K, operating cash flow before investment = $110K.

Depreciation: Why We Add It Back

Depreciation is a **non-cash expense**. It was deducted to compute EBIT (and therefore reduces taxes), but no cash actually left the firm.

The Tax Shield

Depreciation is valuable because it reduces taxes.

If depreciation expense = $40K and tax rate = 30%, the firm pays $12K less in taxes than it would without depreciation.

We capture this by including depreciation in the EBIT calculation, then adding it back to get cash flow.

Straight-Line vs. Accelerated

- **Straight-line:** equal amount each year over useful life
- **MACRS (IRS accelerated):** larger deductions early, smaller later

Accelerated depreciation is generally preferable — larger tax savings sooner.

For simplicity (and for this course), we typically use straight-line.

Capital Expenditures (CAPEX)

What CAPEX Represents

Cash spent on **new property, plant, and equipment**.

Unlike depreciation (which matches historical spending to revenues), CAPEX is a current-period cash outflow.

CAPEX = Δ Net PPE + Depreciation Expense

If net PPE grows from $360K to $440K and depreciation is $40K: CAPEX = $80K.

CAPEX vs. Depreciation

These are **not the same** and should not be confused.

- Depreciation reflects the gradual expense of past spending
- CAPEX is new spending required to sustain or grow the business

A growing firm typically has CAPEX > Depreciation because it spends on new assets faster than old ones are written off.

from FCF)

- **Declining revenues** (project wind-down) → NWC decreases → cash inflow (add to FCF)
- **Project termination** → remaining NWC is

Putting It Together: JC Crawford Enterprises

Scenario: Regional distribution center for Mr. Fix-It-Up franchises in Miami. Initial investment: $500K ($400K equipment + $100K working capital). Revenues: $1M in Year 1, growing 10%/year. 3-year analysis.

	Year 0	Year 1	Year 2	Year 3
EBIT	—	$60,000	$60,000	$60,000
Taxes (30%)	—	(18,000)	(18,000)	(18,000)
NOPAT	—	42,000	42,000	42,000
+ Depreciation	—	40,000	50,000	61,000
− CAPEX	(400,000)	(80,000)	(94,000)	423,000
− Δ NWC	(100,000)	(10,000)	(11,000)	121,000
Free Cash Flow	**(500,000)**	**(8,000)**	**(13,000)**	**647,000**

Note: CAPEX in Year 3 is negative (a cash inflow) because net PPE falls to zero at project end — assets are sold. NWC reverses for the same reason. The terminal year captures the liquidation value.

Building This Model with Claude

What you tell Claude

```
Build an Excel workbook for a 3-year distribution center project. Inputs: $400K equipment
investment (straight-line depreciation, 10-year life), $100K initial working capital, Year 1
revenues of $1M growing 10%/year, gross profit margin 30%, operating expenses 20% of sales
(before depreciation), tax rate 30%, net PPE maintained at 40% of next year's sales, net
working capital at 10% of current year's sales. Create: (1) an Inputs sheet with yellow input
cells, (2) a Pro Forma Income Statement sheet, (3) a Balance Sheet sheet, (4) a Free Cash Flow
sheet showing NOPAT, depreciation add-back, CAPEX, change in NWC, and FCF for Years 0–3, and
(5) a Results sheet with NPV at a 10% discount rate and IRR. Use professional formatting.
```

Claude generates the full workbook — linked formulas across all sheets, proper formatting, and working NPV/IRR calculations. Change any yellow input cell and all outputs update automatically.

Breakout

Sunk Costs and Opportunity Costs

Meridian Pharmaceuticals spent **$15 million** developing a topical pain relief cream. The compound works but performed only modestly better than OTC competitors in clinical trials. The marketing team proposes a national launch, projecting **$8M annual revenue** and **$6M annual operating costs**. Launch requires **$4M in manufacturing equipment** (5-year straight-line depreciation, zero salvage) and **$1.2M in initial working capital**. Tax rate: 30%, WACC: 12%.

Before the launch decision is final, a specialty pharmacy chain offers to **license the compound for $5M**.

In your group (15 minutes)

1. Meridian's CFO insists the $15M R&D must be "recovered" before the project can be approved. Is this correct?
2. Calculate the Year 1 free cash flow for the launch project.
3. The $15M is gone either way — but does it play any role at all in the launch-vs.-license decision?
4. What is the true initial outlay for the launch decision once you account for the license offer? Should

From Assumptions to Cash Flows

Lecion Electronics: A Comprehensive Example

Context

Lecion Electronics has manufactured flat-panel LCDs for computer monitors for 10 years. By 2003, large-screen LCDs (42 inches+) were emerging as the future of home TV.

The proposal: invest **$200 million** in a new fabrication plant in South Korea capable of producing 42-inch LCD panels — making Lecion a major supplier for the home entertainment market over five years (2004–2008).

Strategic Assessment

Before building the financial model, Lecion addressed the key strategic questions:

- **Competitors:** four rivals with equal capability, but Lecion has a 1–2 year technological lead from the dot-com downturn
- **Customer demand:** early adopters are price-insensitive; pricing power will erode over time
- **Complementary products:** Lecion's partnerships with audio manufacturers ensure compatibility
- **Labor:** South Korea location largely insulates from U.S. wage inflation

Market Share

Lecion estimates a **20% market share** of 42-inch LCDs throughout the project's five-year life.

This is subject to significant uncertainty — sensitivity analysis on market share is appropriate.

Price Decay Function

High-tech product prices fall as cumulative volume doubles — the **experience curve**.

Lecion assumes a **20% price decay per doubling** of cumulative market volume:

- 2004: $8,000 per unit (2M units sold by Lecion)
- 2005: $5,959 (market doubles to 15M total)
- 2006: $4,932
- 2007: $4,381
- 2008: $4,098

Variable Cost per Unit

Lecion uses a **70% learning curve model**: each time cumulative production volume doubles, variable cost per unit drops to 70% of its previous level.

- 2004: $9,000/unit (costs exceed price — expected in early ramp)
- 2005: $6,238/unit
- 2006: $4,610/unit
- 2007: $3,815/unit
- 2008: $3,428/unit

The contribution margin turns positive in 2006 as the learning curve brings

Fixed Costs and Depreciation

- Fixed cash operating costs: **$50M/year**
- Plant depreciated straight-line over 5 years: **$40M/year**
- No additional CAPEX or working capital investment assumed after the initial $200M

Simplifying assumptions to keep the model tractable — real analyses would include more detail.

The FCF Estimate: Lecion Electronics

Year	Revenues	Total Expenses	NOPAT	Depreciation	FCF	Contribution Margin
2003	—	—	—	—	$(200,000M)	—
2004	$16,000M	$(18,090M)	$(1,463M)	$40M	$(1,423M)	−12.5%
2005	17,878M	(18,804M)	(648M)	40M	(608M)	−4.7%
2006	19,728M	(18,529M)	839M	40M	879M	+6.5%
2007	17,525M	(15,350M)	1,523M	40M	1,563M	+12.9%
2008	12,294M	(10,375M)	1,343M	40M	1,383M	+16.3%

FCF is negative in 2004–2005 as the learning curve catches up to the experience-curve price decline. The project turns cash-flow positive in 2006 and remains positive through 2008. Tax rate = 30%.

Valuing Investment Cash Flows

Step 3: Discounting

Once FCF is forecast, discounting is mechanical:

$$\text{Present Value of Year } t \text{ FCF} = \frac{\text{FCF}_t}{(1+r)^t}$$

$$\text{Project Intrinsic Value} = \sum_{t=1}^{N} \frac{\text{FCF}_t}{(1+r)^t}$$

$$\text{Net Present Value} = \text{Project Intrinsic Value} - \text{Initial Outlay}$$

The discount rate r reflects the riskiness of the cash flows. For now we treat it as given — Chapters 4 and 5 explain how to estimate WACC for projects of different risk profiles.

NPV: Lecion Electronics

Discount rate: **18%** (Lecion's opportunity cost of capital for this project)

Year	FCF	Discount Factor	Present Value
2003	$(200,000M)	1.0000	$(200,000M)
2004	(1,423M)	0.8475	(1,206M)
2005	(608M)	0.7182	(437M)
2006	879M	0.6086	535M
2007	1,563M	0.5158	806M
2008	1,383M	0.4371	605M
		Project intrinsic value	**$303M**
		Initial outlay	$(200M)
		Net present value	**$103M**

The project creates approximately $103 million of value above its cost — assuming the forecasts are correct. The critical qualifier in every DCF analysis.

IRR: An Alternative Metric

What Is IRR?

The **internal rate of return** is the discount rate that makes NPV = 0.

$$\text{Outlay}_0 = \sum_{t=1}^{N} \frac{\text{FCF}_t}{(1 + \text{IRR})^t}$$

For Lecion: **IRR = 20.24%**

Decision rule: accept if IRR > cost of capital (18%). Lecion's IRR exceeds 18%, consistent with positive NPV.

IRR vs. NPV

When IRR and NPV agree, use either. When they conflict, **NPV wins**.

IRR can mislead when: - Cash flows change sign more than once (multiple IRRs possible) - Comparing mutually exclusive projects of different sizes or timing - Projects have very different lives

IRR is widely used in practice (75% of CFOs report using it), but NPV is theoretically superior.

Multiple IRRs: A Technical Hazard

If a project's cash flows change sign more than once (positive → negative → positive, for example), there can be **multiple IRRs** — all of which make NPV = 0.

Excel and financial calculators will find one IRR without warning you that others exist.

Example

Year 0	Year 1	Year 2	Year 3
$(200)	$1,200	$(2,200)	$1,200

This stream has three IRRs: 0%, 100%, and 200%. The NPV profile oscillates — there is no single "the" IRR.

Solution

When cash flows change sign multiple times, **rely on NPV** rather than IRR.

Build the NPV profile (NPV at various discount rates) to understand the investment's value across a range of assumptions.

The Problem

Often a firm must choose **one option from a set** — different technologies for the same plant, building now vs. building later, acquiring Target A vs. Target B.

In these cases, picking the highest IRR can be wrong. IRR does not account for differences in:

- Project scale (a small high-IRR project may create less total value than a large lower-IRR one)
- Project timing (IRR ignores the reinvestment rate assumption)

The Solution

For mutually exclusive projects, **choose the highest NPV**.

NPV directly measures the dollar value created. It answers the right question: which option makes shareholders wealthiest?

IRR answers a different question: which option offers the highest percentage return? That is not the same thing when projects differ in size.

Breakout

FCF Mechanics

Ridgeway Furniture is evaluating a **$2 million** distribution center (straight-line depreciation over 10 years, zero salvage). Annual cost savings (treated as revenues) and operating expenses are:

	Year 1	Year 2	Year 3
Cost savings	$900,000	$990,000	$1,089,000
Operating expenses	$360,000	$396,000	$436,000

Operating net working capital = **15% of that year's cost savings**, starting Year 1. Tax rate: **30%**. Working capital is fully recovered at Year 3.

In your group (15 minutes)

1. Build the pro forma income statement (EBIT) for each year. Depreciation = $200K/year.
2. Compute FCF for each year: NOPAT + Depreciation − ΔNWC. (No additional CAPEX after the initial investment.)
3. Ridgeway's CFO says "net income is positive but free cash flow is lower — something is wrong with the model." Explain what's happening and identify which line items cause the gap.

Breakout

Cannibalization and Salvage Value

NovaBrew sells **2 million cases of craft beer/year** at a **$4.00 contribution margin per case**. They are evaluating a new craft soda line requiring a **$3M equipment investment** (5-year straight-line depreciation, **$300K salvage value**) plus working capital of **10% of annual soda revenues**.

Soda projections: Year 1 revenues $5M, growing **8%/year** through Year 5. Variable costs: **55% of revenues**. Fixed cash operating costs: **$400K/year**. Tax rate: **30%**, WACC: **11%**.

Warning from the sales team: the soda launch will reduce beer volume by **4%** starting Year 1.

In your group (15 minutes)

1. Compute Year 1 FCF ignoring cannibalization. (Depreciation = ($3M − $300K)/5 = $540K/year)
2. Compute the after-tax cost of cannibalization in Year 1. What is the true incremental FCF?
3. How should the **$300K salvage value** be treated at the end of Year 5? (Book value = $300K, so no taxable gain.)

Summary

Incremental Only

- Only cash flows that change because of the investment belong in the analysis
- Cannibalization of existing products is a real cost
- Sunk costs never belong — they are gone regardless of the decision
- Opportunity costs do belong — they

Expected, Not Hoped-For

- Theory calls for probability-weighted expected cash flows
- In practice, optimism bias systematically inflates forecasts
- When forecasts are optimistic, discount rates must be raised to compensate
- The Phase II review

Project FCF Formula

FCF = NOPAT + Depreciation − CAPEX − ΔNWC

- Add back depreciation (non-cash, but tax shield is real)
- Subtract CAPEX (real cash out)
- Subtract increases in working capital (cash tied up in operations)

Key Takeaways: Valuation

NPV Is the Primary Rule

- Discount FCF at the risk-appropriate rate (WACC)
- NPV > 0: accept — project creates value
- NPV < 0: reject — project destroys value
- For mutually exclusive choices: pick the highest NPV, not the highest IRR

NPV directly measures dollar value creation. IRR does not — it measures percentage return, which is a different (and sometimes misleading) question.

The Story Must Fit the Numbers

A financially attractive NPV built on implausible assumptions is worthless.

- Revenue forecasts must be consistent with the strategic story
- Cost forecasts must reflect realistic learning curves and cost structures
- The sensitivity table reveals which assumptions actually drive the value

The best analysts spend more time on the **assumptions** than on the model mechanics.

Coming Up: Chapter 3

Project Risk Analysis

- Sensitivity analysis: which inputs drive the value?
- Scenario analysis: what happens under different states of the world?
- Break-even analysis: at what level of output does NPV = 0?
- Monte Carlo simulation: modeling the full distribution of outcomes

Why Risk Analysis Matters

A single NPV estimate is almost certainly wrong. The question is not “what is the NPV?” but “what is the range of plausible NPVs, and under what conditions does the project fail?”

Risk analysis transforms a point estimate into a decision-relevant picture of uncertainty.